Multi-Agent Reinforcement Learning for Collaborative Wireless Network Control

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ABSTRACT

The growing complexity of wireless networks, characterized by dense deployments, dynamic traffic, and heterogeneous services, makes traditional centralized control approaches increasingly inefficient. This article proposes a collaborative wireless network control framework based on multi-agent reinforcement learning (MARL), where distributed network entities act as autonomous agents that learn coordinated control policies through interaction with the environment. The proposed approach enables decentralized decision-making while achieving cooperative behavior to optimize network performance metrics such as throughput, latency, energy efficiency, and interference mitigation. Simulation-based evaluations demonstrate that the MARL-based framework offers improved adaptability, scalability, and robustness compared to conventional and single-agent learning methods, making it a promising solution for intelligent control in next-generation wireless communication systems.

1. Introduction

The advent of wireless networks under the beyond 5G and 6G era brings novel difficulties in terms of network size, complexity, and dynamics. Next-generation applications such as massive IoT networks, intelligent transportation systems, and aerial networks require self-governed and adaptive control systems that need to work under uncertain situations. Conventional center-based optimization techniques are not only challenged by scalability but also by adapting to dynamic environmental changes. Hence, multi-agent reinforcement learning (MARL) has received considerable interest in promoting intelligent wireless network control using an innovation-oriented data-driven paradigm [6][33].MARL makes it possible to model wireless network entities like base stations, UAVs, and IRSs as cooperative agents to learn optimal control strategies together by interacting with the wireless network environment. Recently, researchers pointed out the appropriateness of MARL in future 6G wireless networks in terms of scalability, adaptability, and facilitation of self-organizing networks [33][8]. Yet, in MARL environments, a complication is involved in the learning process by virtue of non-stationarity, partial observability, and coordination overhead, which makes its deployment in a wireless network challenging.However, to overcome these challenges, recent literature has concentrated on improving the functionality of MARL by incorporating the concept of emergent communication and offline learning techniques. The study presented in [6] illustrates the effectiveness of learnable communication among the agents in increasing the efficiency of coordination in dense wireless networks. At the same time, the idea of offline learning in the context of MARL has also been explored [17][8] to address the uncertainty and instability of the online learning process.MARL-based solutions have also been used in a variety of wireless network issues in order to validate their efficacy. For instance, cooperative computation offloading and routing in a Cloud Edge network [29] and distributed resource management in a Vehicle and Sensor network [35] illustrate how cooperative decision-making by multiple agents leads to effective results. Additionally, MARL-based control solutions in UAV and RIS-based wireless networks achieved efficacy in coverage and energy efficiency [21][7].Despite these advances, many of the existing wireless control solutions based on MARL focus either on isolated problem settings or rely on simplifying assumptions that limit their general applicability. Many challenges regarding scalability, robustness under highly dynamic network conditions, and effective coordination among heterogeneous agents remain largely open. Motivated by these gaps, this work focuses on Multi-Agent Reinforcement Learning for Collaborative Wireless Network Control, developing from foundational studies [6][33][8] and selective application-driven research [29][35] to explore scalable and adaptive control strategies for next-generation wireless networks.

1.1. Contribution Of This Work

This work presents a collaborative wireless network control framework based on multi-agent reinforcement learning, where distributed network entities learn to coordinate their control decisions in a decentralized manner. The wireless control problem is modeled as a cooperative multi-agent decision-making process, enabling agents to jointly optimize network performance under dynamic and partially observable conditions. The proposed approach enhances scalability and adaptability in dense wireless environments by reducing reliance on centralized control. Through simulation-based evaluation, the effectiveness of the collaborative learning

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framework is demonstrated in terms of improved throughput, reduced interference, and better overall network efficiency compared to conventional and single-agent learning methods.

1.2. Organization of The Article

The remainder of this article is organized as follows. Section II reviews the related work on multi-agent reinforcement learning and its applications in wireless network control. Section III describes the system model and problem formulation, including the cooperative multi-agent framework adopted in this study. Section IV presents the proposed multi-agent reinforcement learning methodology and learning architecture. Section V discusses the simulation setup and performance evaluation results, along with comparative analysis against baseline approaches. Finally, Section VI concludes the article by summarizing the key findings and outlining potential directions for future research.

2. Research Objectives

The main goal of this work is to develop a MARL-based architecture for collaborative control of wireless networks, where multiple entities in the network (e.g., BSs, UAVs, ISs) can individually learn while still executing coordinated control actions. Unlike previous works that mainly emphasize centralized approaches to wireless network control, this study aims to capture and exploit the underlying interlinking among wireless networks via the proposed MARL framework, with the aim of jointly optimizing the overall network performance.

The second goal is to cast the wireless network control task as a cooperative multi-agent Markov decision process (MDP), in which each agent operates with partial observability and acquires coordinated policies via shared or otherwise aligned reward structures. This objective emphasizes enabling effective collaboration among agents for key network control functions—such as resource allocation, interference mitigation, power regulation, and spectrum coordination—while maintaining scalability in dense and heterogeneous wireless environments.

The third goal of this work is to design and assess appropriate multi-agent deep reinforcement learning algorithms that achieve a trade-off between coordination efficiency, learning robustness, and computational cost. By exploiting cooperative MARL approaches, the research aims to improve responsiveness to time-varying traffic loads, user mobility, and channel uncertainty, thereby overcoming the drawbacks of traditional optimization-based schemes and single-agent learning strategies in wireless network control.

The ultimate goal is to evaluate the efficiency of the combined collaborative MARL-based control architecture in extensive performance analysis against the existing and single-agent models. The Fundamental performance measurement factors include reviewing the network throughput, degree of interference reduction, energy efficiency gain, and quality-of-service improvement to prove the efficiency of collaborative machine-based intelligence in next-generation and beyond-5G/6G communication.

2.1. Additional Objectives

The main objective of this research is to develop a collaborative learning-based approach in which multiple wireless network components, such as base stations or access points, learn to work together using multi-agent reinforcement learning. Instead of relying on a centralized controller, each network node learns from its own interactions with the environment and cooperates with other nodes to make effective control decisions.By effectively managing resources, lowering energy consumption in dynamically changing network conditions, and minimizing interference and delay, this method seeks to strengthen overall wireless network performance.

The second goal of this study is to look at how well the proposed multi-agent reinforcement learning framework works and how well it adapts to real-world wireless network situations. This entails examining the system's response to fluctuations in user density, traffic load, and channel conditions. The research aims to illustrate the potential of collaborative learning for scalable, reliable, and intelligent wireless network control in next-generation communication systems by comparing the proposed approach with established conventional and learning-based methods.

The third objective of this research is to create a multi-agent reinforcement learning framework that allows collaborative control in wireless networks. These networks operate under dynamic and distributed conditions. In this framework, multiple network entities, such as base stations, unmanned aerial vehicles, or wireless nodes, act as intelligent agents. They learn to make control decisions based on local observations while working together to improve overall network performance. This goal aims to lessen reliance on centralized control and address the drawbacks of single-agent methods in complex and large-scale wireless environments.

The fourth objective of this research is to investigate and apply appropriate multi-agent reinforcement learning algorithms that can achieve stable and scalable learning in collaborative wireless network control. In particular, the study explores learning strategies that support decentralized decision-making while maintaining coordination among agents during training. This objective also includes analyzing learning convergence, scalability, and communication overhead, as well as comparing the proposed MARL-based approach with traditional optimization methods and independent learning techniques.

The final goal is to evaluate the performance and resilience of the proposed collaborative MARL framework in real wireless network environments. This research will use simulation experiments to measure key performance metrics such as throughput, latency, energy efficiency, and fairness. The results will show how effective and flexible collaborative multi-agent learning is for next-generation wireless

Table 1
Literature Survey on Multi-Agent Reinforcement Learning for Collaborative Wireless Network Control

Ref	Focus Area	Methodology	Key Contributions	Limitations
[3]	Multi-UAV Wireless Networks	Survey of RL and MARL techniques	Comprehensive review of RL-based control for UAV wireless networks	No experimental validation
[25]	UAV-MEC Broadcasting	Adaptive gossip-based reinforcement learning	Improves broadcast efficiency and adaptability in UAV-MEC systems	Limited to broadcast scenarios
[26]	UAV Deployment Optimization	Multi-agent reinforcement learning	Enhances coverage and connectivity via optimized UAV placement	Scalability not fully explored
[30]	Collaborative Wireless Network Control	Deep multi-agent reinforcement learning	Demonstrates effective cooperation among wireless agents	High training complexity
[27]	Vehicular Wireless Networks	MARL-based transmission optimization	Improves throughput under high-mobility conditions	Assumes ideal communication links
[11]	Future Internet Wireless Networks	Distributed multi-agent reinforcement learning	Enables adaptive and decentralized network control in dynamic environments	Performance depends on reward design
[35]	UAV-aided IoT Energy Transfer	Graph-based multi-agent reinforcement learning	Ensures fair and efficient wireless energy delivery	Centralized training overhead
[28]	Next-Generation Wireless Systems	Cooperative MARL-based control framework	Improves adaptability in highly dynamic environments	Limited real-world validation
[13]	Wireless Sensor Networks	Attention-based multi-agent actor–critic learning	Extends network lifetime through joint scheduling	Computationally intensive
[9]	Distributed Medium Access Control	Fully distributed multi-agent reinforcement learning	Achieves fair and efficient channel access	Simplified physical-layer assumptions
[18]	Intelligent Wireless Network Optimization	Multi-agent deep reinforcement learning	Improves network performance through cooperative learning and adaptive control	Increased computational complexity

networks. This includes scenarios with dense deployments, IoT systems, and UAV-assisted communications.

3. Literature Survey

The study [29]introduces a multi-agent deep reinforcement learning (MARL)-based framework, which is an opportunity of a multi agent deep reinforcement learning (MARL) for cooperative computing offloading and route optimization of applications on multiple cloud edge networks. The agents act independently but in collaboration to minimize network latency and save energy from the user devices. This research illustrates how distributed problem solving can be accomplished with MARL under dynamic network conditions and demonstrates how MARL may be applied to the control of wireless collaborative networks.

The study referred to in [22] investigates the application of multi-agent reinforcement learning (MARL) for the dynamic management of wireless resource allocation. Multiple agents are employed to learn how to coordinate their decisions regarding spectrum access and power allocation using a model of the problem as a cooperative Markov Decision Process (MDP) to demonstrate that MARL-based approaches outperform both traditional optimization and single-medium reinforcement learning methods. This is particularly evident in scenarios where networks are dense, and therefore have higher levels of interference, partial observability, etc.

The authors' paper [36] examines the use of multi-agent Reinforcement Learning methods to optimise Wireless Networks, where scalability and learning stability are paramount considerations. Among various challenges that occur in Multi-Agent Systems the authors identify Non-Stationarity and Coordination Overhead and propose modifications to algorithms to enhance Convergence and Robustness. Overall results clearly demonstrate that cooperative learning can provide a significant improvement in Network Throughput and fairness, over independent learning methods.

In [35], MARL has been applied to solve resource allocation problems in wireless networks by using a group of agents who work together to make decisions regarding transmission power, channel selection, and scheduling. The proposed method has implemented a centralized training

structure. However, when using centralization for the training of agents, decentralized execution occurs so that coordination among all agents is accomplished via the centralized structure. The results show that MARL is very effective for use in managing wireless network control problems in real life.

In [4], the authors examine MARL-based solutions for next-generation wireless systems. They highlight the need for flexibility and broad applicability across different network setups. The paper shows that cooperative learning helps agents react well to changing traffic patterns and network structures. This makes MARL a solid option for smart control in 5G and upcoming 6G networks.

The authors of [15] research multi-agent reinforcement learning under conditions of multiple types of wireless devices that are enabled with sensors and internet of things [IoT]. The research demonstrates that by leveraging a cooperative approach to learning from experiences, learning agents will greatly increase energy efficiency and maximize the lifetime of the wireless network. Furthermore, the results of the study support earlier findings regarding the advantages of collaborating between agents engaged in a learning process.

The authors in [10] show how the use of Multi-Agent Reinforcement Learning (MARL) allows for Multihop Wireless Networks consisting of a mix of devices with rapidly changing channels. They emphasize that teams can be built based on cooperation and common learning goals to control and stabilize networks in the long run. They also provide empirical evidence showing how to use the collaborative approach in MARL for effective management of interference and quality of service.

The final work [20] proposes a viable multi-agent reinforcement learning framework to develop generalized wireless MAC communication protocols. This work also provides a solutions to the real-world problems of communication overhead and bad generalization often experienced in previously developed MARL-based MAC Protocols. The method developed from this paper is able to provide strong generalization and performance on a multitude of networks by combining both observation abstraction and decentralized learning. This should make it suitable for use within future applications of collaborative wireless network control.

[19]: The power control scheme for cell-free XL-MIMO systems using multi-agent reinforcement learning is proposed by the authors. The key contribution of their work is to show that the distributed agents can collaborate to optimally achieve the transmit power adaptation over dynamic conditions and can attain significant improvement in terms of spectral efficiency and interference mitigation. This study has demonstrated the suitability of MARL in handling high-dimensional and complex wireless control problems.

[31] has created a new collaborative approach for developing edge cache strategies using federated learning in conjunction with each small base station being viewed as an information processing unit. This creates a group of intelligent agents that will jointly select the best option or strategy for how to store or cache incoming data to optimize both bandwidth and the amount of time to deliver that data (latency). In the test scenarios the results demonstrated that the collaborative decision-making between agents resulted in much greater levels of overall network efficiency which demonstrates the significance of using collaboration in the design of decentralized systems.

The authors of [16] demonstrate a cooperative approach to the joint optimization of UAV trajectories, task offloading, and resource allocation using a Multi-Agent Deep Reinforcement Learning (MARL) methodology applied to UAV-assisted mobile edge computing networks under partial observability. The proposed method allows agents to communicate via exchanging messages, resulting in higher energy efficiency and increased overall network performance. Specifically, this study illustrates how MARL is effective for managing mobility, uncertainty, and dynamically changing topologies in wireless networks.

[32] also evaluates how MARL can impact distributed Channel Access over industrial Edge cloud environments. Based upon extensive research into MARL systems the authors have developed a system for centralised training with multiple agents executing their actions distributedly when choosing access points and channels to communicate. The results show that MARL based systems achieve higher levels of stability and throughput during simulations versus conventional systems; suggesting that these systems hold a strong potential for providing a scalable and dependable means of providing wireless network control for mission critical applications.

[14] develops an application for MEDAC using multiple agent deep reinforcement learning. By applying a cooperative approach of MARL the authors demonstrate that through the use of cooperative learning networks, nodes within the network can work collaboratively to determine the optimum window parameter settings for contention, thus maximising throughput and minimising packet collisions based upon the conditions that are going to be taken into account during the course of their joint operation. The findings of this paper also support that MARL based systems provide superior-performance over both non-cooperative and single agent MARL systems for the purpose of decentralised optimisation of wireless performance under either static or dynamic conditions.

The use of graph-based communication models with Graph Neural Networks (GNNs) was presented in [2] to solve convergence and coordination issues in MARL via distributed power control in wireless systems. The graph-based technique aids in providing stability, scalability, and generalization during the agent-to-agent interaction (adaptively via Adaptive Graphs) improving how agents will learn and evolve over time. This research shows the importance of having a structured collaboration and sharing of resources between multiple agents in the control of Wireless Communications.

[12] used MARL to solve User Association and Interference management within UAV-assisted Cellular Networks

and Rate-splitting Multiple Access Networks by converting the system's model into a partially observable Markov Decision Process, allowing UAVs to optimize for collaborative uplink downlink associations and collaborated Beamforming strategies. Results demonstrated an improvement in Spectral Efficiency and a reduction in Interference, proving the effectiveness of MARL in complex scenarios with Multiple Agent Interactions.

The last report, [1], offered a novel multi-agent Q-Learning framework for Real time load balancing, User Association, and handover management in Dense Mobile Networks. By treating each user as an Autonomous Agent, the authors could adapt their behaviour towards their channel variations and mobility while providing base stations with the load constraints required. This approach offered faster convergence time and higher throughput rates than previously shown through standard methods, confirming that MARL has a superior ability to perform Adaptive and Collaborative Control Functions of Wireless Communications.

Cognitive Radio Networks (CRNs) are looking to improve their overall performance through using Multiple-Agent Reinforcement Learning (MARL) methods. MARL allows network users (agents) to cooperatively sense and share spectrum resources while reducing interference to Primary Users. Additionally, [24] focuses on the cognitive radio technology, describing difficulties related to the Convergence Issues and the Scalability of Solution Space, and Limited Computational Resources. All three factors are essential to consider for the design of Cooperative Wireless Networks.

Boutiba et al. [5] explore the application of multi-agent deep reinforcement learning for dynamic Time Division Duplexing (TDD) in multi-cell 5G networks, building on the idea of cooperative decision-making. Their work tackles the challenge of minimising cross-link interference between adjacent cells while adjusting uplink and downlink configurations in response to changing traffic demands. The suggested method outperforms static and single-agent solutions in terms of throughput, buffer overflow, and interference management by utilising a fully decentralised MARL framework based on centralised training and decentralised execution. The efficiency of collaborative MARL in handling challenging resource allocation issues in next-generation wireless networks is confirmed by this study.

Pham et al. in [23] contribute to a growing body of literature examining Distributed Cognitive Radio Networks and describe a distributed cognitive radio network architecture using Directional Antennas to determine the optimal energy sensing threshold as a function of Spectrum and Synchronization. The authors create a solution for decentralized sensing of Spectrum in a MAC environment, using a decentralized partially observable Markov Decision Process to model and solve the associated optimization. The authors provide a method for Cooperative Learning by using multiple Autonomous Mobile Device Learning Policies (MADDPG). The results of the proposed solution show significant increases in accuracy of Detector performance, increased Speed of Convergence, and significant decreases in False Alarm rates compared to baseline models. The work demonstrates how MARL-based Learning can successfully solve Partial Observability and Uncertainty Challenges associated with Wireless environments and how using a Decentralized approach may create the potential for developing Collaborative Decision-making Processes.

Another important issue that MARL techniques address is energy efficiency in large-scale wireless networks. A multi-agent reinforcement learning framework is proposed by Zhao et al. [34] to maximise energy efficiency in LoRaWAN-based wireless underground sensor networks. Each sensor node in their method functions as an independent agent that modifies its transmission parameters in response to local factors like packet collisions, energy consumption, and link quality. By increasing network lifetime and adaptability in dynamic subterranean environments, the suggested MARL-based solution performs better than conventional adaptive data rate mechanisms. The potential of MARL for cooperative and energy-conscious control in wireless networks with limited resources is highlighted in this study.

3.1. Research Gaps(Discussion of Base papers

Despite recent progress in using multi-agent reinforcement learning for wireless network control, current work still has clear gaps. Many approaches depend on centralized training or assume access to full network information, which makes them hard to scale and deploy in real systems. Most studies also focus on narrow tasks like resource allocation or routing, instead of coordinating control across the whole network. Challenges such as stable learning, coordination with only partial information, and adapting to changing traffic and channel conditions remain underexplored. There is also limited attention to making these methods generalize across different types of networks, highlighting the need for more robust and scalable collaborative MARL solutions.

3.2. Motivation

Wireless networks are growing fast and becoming more complex, with many closely spaced devices, different types of services, and constantly changing conditions. Old, centralized control methods can't keep up with shifting traffic, moving users, and variable signal quality. Learning-based approaches help, but a single controller can't capture the cooperation needed across many distributed network nodes. Multi-agent reinforcement learning (MARL) offers a promising solution by letting multiple network nodes learn to coordinate their actions. This work explores collaborative MARL-based control to make future wireless networks more scalable, adaptive, and resilient.

3.3. Problem Statement

Modern wireless networks are inherently dynamic and complex, rendering centralized and static control schemes ineffective. Although reinforcement learning provides adaptability, single-agent methods are inadequate for addressing the cooperative behavior required in distributed wireless

environments. Current multi-agent approaches, however, struggle with scalability and effective coordination. Consequently, there is a need for a collaborative and adaptive multi-agent reinforcement learning framework that can support efficient, decentralized control of wireless networks operating under changing conditions.

4. Conclusion

EFS-IDS is an efficient, scalable intrusion detection framework that can be used to overcome the shortcomings of the traditional IDS in an IoT context due to the problem of class imbalance.

Declarations

Ethical Approval

This manuscript reports studies that do not involve human participants, human data, human tissue, or animals.

Conflict of Interest

The authors have no conflict of interest to declare that are relevant to the content of this article.

Authors' Contributions

S. Gopikrishnan contributed to the conceptualization, Formal analysis, drafting the original manuscript, and designing the experimental protocols. Author-2 was responsible for Conceptualization, methodology, software implementation, and dataset curation. Author3 contributed to conceptualization, supervision, and in-depth analysis of the experimental results. Author-3 handled the review and editing of the final draft, as well as the overall validation of the study results.

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Availability of data and materials

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